Alex Chen

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Education

Bachelor of Applied Science, Computer Science *Sept. 2022 – Apr. 2026*
McMaster University (cGPA: 3.92 / 4.00) Hamilton, ON

- **Notable Coursework**: Data Structures and Algorithms, Databases, Intro to Data Science Methods, Statistical Methods and Applications, Intro to Machine Learning, Operating Systems, Applied Cryptography
- **Awards**: McMaster Engineering Award of Excellence

Technical Skills

Languages: Python, Javascript, Typescript, HTML, CSS, Java, R, SQL, Haskell, C, Bash, SAS, Elm
Libraries: React, PyTorch, Tensorflow, Numpy, Pandas, React Spring, BeautifulSoup4, Selenium, Pydub
Developer Tools & IDEs: Figma, Git/Github, Visual Studio Code, RStudio, SQLite3, Netbeans
APIs & Frameworks: Tailwind CSS, Bootstrap, Node.js, Express.js, Flask, SQLAlchemy, React Native, Docker, Pytest
Methodologies: Agile, Scrum, Kanban Boards
Other Tools: Unified Modelling Language (UML), Microsoft Office, MATLAB, Linux (Ubuntu), Google Suite

Work Experience

Google Developer Student Club McMaster *Sept. 2023 – Apr. 2025*

- Currently one of 20+ members on GDSC's open source team, developing open-source projects through weekly sprints in an **Agile** development environment
- Contributed to the front-end, back-end, and database aspects of web projects throughout the entirety of their 14-week development cycles, from planning to deployment

Music Recital Piano Accompanist *Mar. 2020 – Apr. 2020*

- Coordinated the performances of 10+ music students for a recital held by the University Settlement of Toronto
- Scheduled over 20 rehearsal sessions in total for the performers to ensure all musical pieces were polished, which resulted in a cohesive performance praised by audiences

Projects

Ocular Disease Identifier | *Typescript, Python, HTML, CSS* *Oct. 2024 – Apr. 2025*

- Helped to develop the frontend and ML model for a web app to help ophthalmologists quickly identify ocular diseases in eye scans for further examination
- Website uses a **Tensorflow** convolutional neural network that takes fundus images as input, then classifies it as one of many eye conditions based off a preclassified dataset
- Designed the UI and user flow in **Figma**, then implemented the design with **React.js**, **TypeScript**, & **Tailwind**
- Implemented ML **data pipelines** to merge and preprocess multiple image datasets into one **Pandas dataframe**

Gamified Learning Platform | *Javascript, Python, HTML, CSS, SQL* *Sept. 2023 – Apr. 2024*

- Co-produced an educational platform website to help students learn in a productive and enjoyable manner
- Platform is designed to tackle students' distracted learning and low engagement rates by adding interactive game-like elements to education modules (e.g. leaderboards, achievements, point-redemption systems)
- Implemented a responsive front-end and a **REST API** server back-end with **HTML**, **CSS**, **JS**, **Jinja**, & **Flask**
- Utilized Python's **SQLAlchemy** to create a **relational database** for user data and educational modules, and **designed API endpoints** to access and modify the database from the front-end
- Wrote **unit** & **integration tests** in **Pytest** to verify the functionality of the front-end, back-end and API calls

BusBuddy: Route Optimization Tool | *Javascript, Python, HTML, CSS, SQL* *Jan. 2024 – Jan. 2024*

- Co-developed a website tool prototype for the 24-hour **hackathon Deltahacks X**, which aims to give more students access to school transportation and commute to school as efficiently as possible
- Project won **1st place** for "Best DEI (Diversity, Equity, & Inclusion) Hack"
- App utilizes a clustering algorithm on students' addresses to formulate optimal bus stop locations in an area, then using depth-first search, finds the shortest fuel-efficient path between stops for bus drivers to follow
- Website uses **Google Maps API** to generate bus stops and routes from inputted coordinates, and manages student addresses within a **relational database** with **SQLAlchemy**
- Implemented the front-end from **Figma mockups**, while hooking up the authentication system to the website's interface and constructing API endpoints in the back-end **Flask** server